

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 525 — Past Questions & Materials

by Renance

Course: EEE 525

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 500

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Paper 1 of 2 — EEE 525 - Mechatronics II

Level 500 • Second Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY**

EXAMINATION: Second Semester
SESSION: 2025/2024
DATE: 31/10/2025



COURSE CODE: EEE 525
COURSE TITLE: Mechatronics II
TIME ALLOWED: 2h,30mins

INSTRUCTION: ANSWER QUESTION ONE, THREE ANY OTHER QUESTION

Q1 (a) Mention two merits and demerits of potentiometer. (4marks)

(b) A linear resistance potentiometer is 200 mm long and is uniformly wound with a wire of total resistance 0.005 MΩ. Under normal conditions, the slider is at the centre of the potentiometer. Determine the linear displacement when the resistance of the potentiometer measured by wheatstone bridge is 0.0000015 GΩ. If it is possible to measure a minimum value of 5 Ω resistance with the above arrangement, determine the resolution of the potentiometer in metres. (6marks)

(c) An LVDT is employed for measuring the Deflection of a bellow. The sensitivity of the LVDT is 600 V/mm. The bellow is deflected by 0.155 mm by a pressure of 1.2×10^6 N/m². Determine the sensitivity of the LVDT in Volts per N/m² and the pressure when the output voltage is 4.5 V. (4marks)

(d) A pump supplies oil at 126 LPM to a 6 cm diameter double-acting hydraulic cylinder. If the load is 0.00001 GN (extending and retracting) and the rod diameter is 3 cm, find

- (i) Hydraulic pressure during the extending stroke. (2marks)
- (ii) Piston velocity during the extending stroke. (2marks)
- (iii) Cylinder power in Watts during the extending stroke. (2marks)
- (iv) Hydraulic pressure during the retracting stroke. (2marks)
- (v) Piston velocity during the retracting stroke. (2marks)

Q2. a(i) List six advantages of Programmable Logic Controller (PLC) over conventional relay (3marks).

(ii) State six factors to consider when choosing PLCs. (3 marks)

(b) Draw the ladder diagram of the following of the following boolean expression. (3marks)

(i) $Y = (\overline{A}\overline{B} + ABC) \cdot D$ (3marks)

(ii) $Y = B [C + (\overline{D}\overline{E}\overline{C}) + \overline{F}C]$ (4marks)

(iii) $Y = (\overline{A}\overline{B}C + A\overline{B}\overline{C}\overline{D} + \overline{A}BCD + \overline{A}\overline{B}CD) + D$ (5marks)

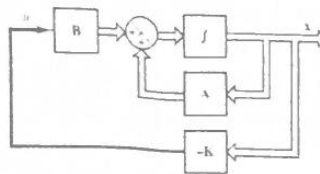
(iv) $Y = A \oplus B \oplus C$ (4marks)

Q3 (i) A system is describe by the expression $x' = Ax + Bu$

$$\text{Where } A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The system uses the feedback control $u = -Kx$. Our desired closed loop pools are $\mu_1 = -2 + j3$, $\mu_2 = -2 - j3$, and $\mu_3 = -5$. Determine the state feedback gain matrix K using two different pole placement techniques and compare the results. (14marks)

3 (ii) Deliberate the effect of controllability and observability of the system below. (3marks)

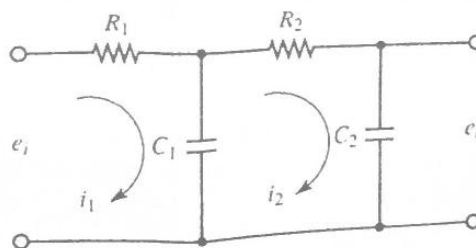


3(iii) A system controlled with PID shows oscillations. What could be the reason, and how would you adjust the controller parameters? (7marks)

4(i) Expatriate on the different classification of a control system and distinguish between open-loop and closed-loop systems with examples. (4marks)

4(ii) Explain how feedback control improves precision in CNC machines. (3marks)

4(iii) Find the transfer function of the system shown below. (10marks)



4(iv) Show a comparative analysis between pneumatic, hydraulic, electrical and mechanical systems and outline their shortfalls and standouts. (5marks)

GOODLUCK

Paper 2 of 2 — EEE 525 MECHATRONICS 11 SECOND SEMESTER 2022/2023

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MARINE ENGINEERING
RUNNING AND MAINTENANCE OF SHIP POWER PLANT (MAR 521)

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QUESTION ONE ¶

Two MAN B & W 6-cylinder inline Main Engine was newly installed in the port and starboard side of the machinery space in a container ship.

- A. Discuss the required performance from the two main Engine 1 mark
- B. Develop a maintenance concept that guarantee's optimal performance of the Two Main Engine 2 marks
- C. If the working fluid for the Main Engine is diesel oil (DO), discuss how the working cycle is achieved with respect to strokes of an engine if the Engine is classified as a four-stroke engine 4 marks
- D. If functional failure occurs in the portside main engine and clear instructions has been given on not engaging the starboard main engine, what can be done and what are the key components that can drive such process.? 4 marks
- E. If an alarm is raised from the bridge due to excessive black smoke exiting through the Port side main engine funnel, what are the likely causes and preventive measures ' 4 marks
- F. How can we achieve a clean machinery work-space that can prolong the main engines lifespan? 3 marks

QUESTION TWO

- A. Explain two reasons for the vast utilizations of internal combustion Engines in Africa over external combustion engine 2 marks
- B. Discuss the associated challenges of running internal combustion engines with fossil fuels 1 mark
- C. List five key significance for the popularity of Diesel engine 5 marks
- D. Describe two stroke Engines and state the relevance of scavenging. 2 marks
- E. State the difference between spark ignition and compression ignition engine with respect to working fluid, ignition source'and working cycle 2 marks

CLEMENT A.I

MNIMENA